

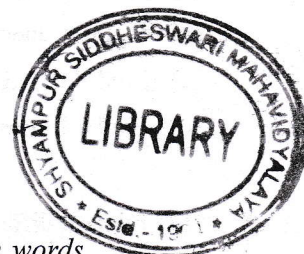
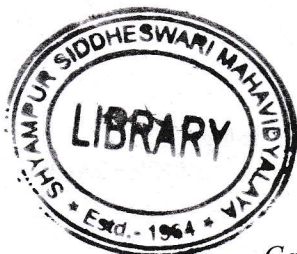
2022

COMPUTER SCIENCE — GENERAL

Paper : DSE-A-1

(Database Management System)

Full Marks : 50



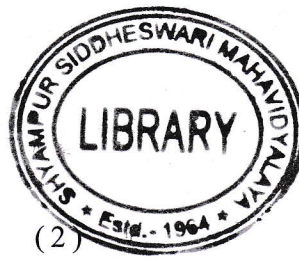
The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Answer **question no. 1** and **any four** questions from the rest.

1. Answer **any five** questions of the following : 2×5
 - (a) What do you mean by data redundancy?
 - (b) Define weak entity with example.
 - (c) Explain derived attribute and multivalued attribute with example.
 - (d) What is meant by full functional dependency?
 - (e) Define candidate key with example.
 - (f) Explain cartesian product operation with example.
 - (g) Explain Transitive dependency.
 - (h) Why is Normalization required?
2.
 - (a) Define metadata.
 - (b) Discuss about any three types of data models of DBMS. 1+(3+3+3)
3.
 - (a) What are the components of ER diagram?
 - (b) What are the different types of relationship that can be present in an ER model?
 - (c) Construct an ER diagram to store data in a library of your college. 3+3+4
4.
 - (a) Give an example of unnormalized form of a relation R. Hence, explain the importance of normalization with proper example.
 - (b) Explain 2NF and 3NF with examples. (2+2)+(3+3)
5. Given the following relational schema :
Book (BookId, Bname, Title, Author, Subject, Price, Availability)
Borrower (Bid, Bname)

Please Turn Over



X(5th Sm.)-Computer Sc.-G/DSE-A-1/CBCS

(2)

Borrows (*BookId*, *Bid*, *DateOfIssue*, *DateOfReturn*, *Fine*)

Perform the following :

- (a) Write a SQL statement to insert a tuple in Borrows table.
 - (b) Write a query in relational algebra to display names of all Borrowers.
 - (c) Find the details of the highest priced book using SQL.
 - (d) Find all borrowers who have been fined between 01.11.2022 to 31.12.2022 using SQL.
 $2\frac{1}{2}+2\frac{1}{2}+2\frac{1}{2}+2\frac{1}{2}$
6. (a) Explain different types of 'Join' operation of relational algebra.
- (b) Explain with suitable example, Specialization and Generalization. 6+(2+2)
7. (a) Explain with diagram, Three-Schema Architecture of DBMS.
- (b) Differentiate between logical data independence and physical data independence.
- (c) 'SQL is a relationally complete language'. Justify. 4+4+2
8. (a) Consider a relation R with five attributes A, B, C, D, E having following dependencies :
 $A \rightarrow B$, $BC \rightarrow E$ and $ED \rightarrow A$
- (i) List all keys for R
- (ii) In which normal form the table is? Justify your answer.
- (b) Define tuple and cardinality of a relation. (2+4)+4
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